LinkedIn edunet foundation

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Unlocking Sales Insights for Revenue Growth

Project Domain

Excel

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT

This project focuses on cleaning and analyzing a retail sales dataset and creating visual dashboards to generate meaningful business insights. The dataset contained sales transaction records including product category, region, quantity, price, discount, and profit.

The raw dataset was cleaned by checking for duplicate records, correcting data formats, and validating numerical values. Microsoft Excel were used to analyze the dataset and build visual dashboards such as sales trends, profit by product category, and regional sales performance.

The final output of this project includes a cleaned dataset and interactive visualizations that help understand business performance. This project demonstrates how data analytics tools can transform raw sales data into useful insights for decision-making.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project i.e. Unlocking sales insights for revenue growth

1.1 Problem Statement

A retail organization might receive sales information from various operational systems like

- Inconsistent date format
- Incorrect or manually computed totals
- Region and category inconsistencies
- Limited visibility into sales trends and performance

Due to such issues, management is having a hard time keeping an accurate record of sales revenue, determining high and low-performing products, and assessing the sales regions' performance.
"The objective of the project is to clean and standardize the sales data, as well as create dashboards to ensure the accuracy of the sales insights."

1.2 Motivation

The motivation behind this project is to understand how raw business data can be transformed into useful information through data cleaning and visualization. Data analytics is an important skill in modern industries, and tools like Excel are widely used for business intelligence.

1.3 Objectives

- Understand the structure of the retail sales dataset.
- Clean the dataset by removing duplicates and fixing errors.
- Handle missing values and inconsistent data formats.
- Analyse sales data using Excel.
- Create visual dashboards to show insights.

1.4 Scope of the Project

This project focuses on cleaning and analysing retail sales data using Microsoft Excel. The project includes data preprocessing, analysis, and dashboard creation. Advanced machine learning models or real-time data systems are outside the scope of this project.

CHAPTER 2

Literature Survey

2.1 Existing Work and Research

Many organizations use data analytics tools to understand business performance. Tools like Excel are commonly used to clean datasets and create dashboards.

Research in business intelligence shows that visualization tools help organizations identify patterns such as sales trends, customer behaviour, and product demand.

2.2 Tools and Technologies Review

Microsoft Excel

Used for:

- Data cleaning
- Removing duplicates
- Using formulas
- Pivot tables

2.3 Gaps Identified

- Many datasets contain errors and inconsistencies.
- Manual data cleaning takes time.
- Visual dashboards are needed for better decision-making.
- This project addresses these issues by combining data cleaning and visualization techniques.

CHAPTER 3

Proposed Methodology

3.1 System Design

The project follows these steps:

1. Import dataset into Excel
2. Clean the dataset
3. Perform analysis
4. Create visual dashboards
5. Generate insights

Flow:

Raw Data → Data Cleaning → Data Analysis → Visualization → Insights

3.2 Modules / Components Used

Module 1 – Data Ingestion

Import dataset into Excel.

Module 2 – Data Cleaning

- Remove duplicates
- Fix date formats
- Check missing values

Module 3 – Data Analysis

- Use pivot tables
- Calculate totals and averages

Module 4 – Visualization

Create charts and dashboards.

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0:

Raw Sales Data → Data Cleaning System → Dashboard Output

DFD Level 1:

Raw Data
↓
Data Validation
↓
Data Cleaning
↓
Data Analysis
↓
Visualization

3.4 Advantages

- Improves data accuracy
- Helps identify business trends
- Saves time in analysis
- Provides easy-to-understand visual insights

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher

Table 1

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365

Table 2

CHAPTER 4

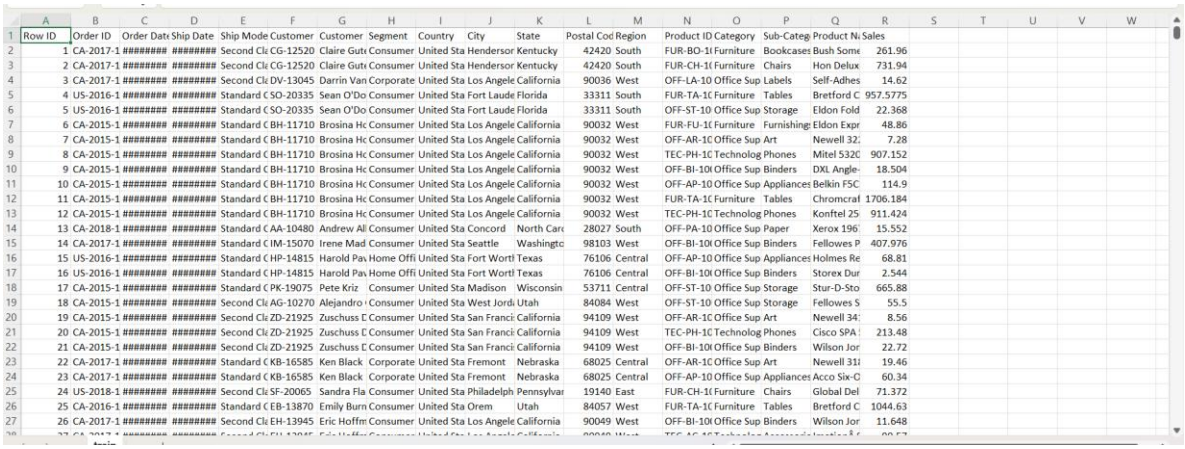
Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

Describe the data cleaning process and present the before/after comparison of your dataset.

- Dataset source: Dataset is downloaded from Kaggle
- Original dataset size: Number of rows-1001; columns in the raw data-10
- Issues found: duplicates, null values, wrong data types, formatting inconsistencies,
- Cleaning steps applied: Removing the duplicates from data, checking the wrong data types.
- Cleaned dataset size: number of rows after cleaning- 1001; number of columns after cleaning- 15



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer	Segment	Country	City	State	Postal Cod	Region	Product ID	Category	Sub-Categ	Product Na	Sales
2	1	CA-2017-1	########	########	Second Cla	CG-12520	Claire Gut	Consumer	United Sta	Hendersor	Kentucky	42420	South	FUR-BO-1(	Furniture	Bookcases	Bush Some	261.96
3	2	CA-2017-1	########	########	Second Cla	CG-12520	Claire Gut	Consumer	United Sta	Hendersor	Kentucky	42420	South	FUR-CH-1(	Furniture	Chairs	Hon Delux	731.94
4	3	CA-2017-1	########	########	Second Cla	DV-13045	Darrin Van	Corporate	United Sta	Los Angele	California	90036	West	OFF-LA-10	Office Sup	Labels	Self-Adhes	14.62
5	4	US-2016-1	########	########	Standard C	SO-20335	Sean O'Do	Consumer	United Sta	Fort Laude	Florida	33311	South	FUR-TA-1(	Furniture	Tables	Bretford C	957.5775
6	5	US-2016-1	########	########	Standard C	SO-20335	Sean O'Do	Consumer	United Sta	Fort Laude	Florida	33311	South	OFF-ST-10	Office Sup	Storage	Eldon Fold	22.368
7	6	CA-2015-1	########	########	Standard C	BH-11710	Brosina Ho	Consumer	United Sta	Los Angele	California	90032	West	FUR-FU-1(	Furniture	Furnishing	Eldon Expr	48.86
8	7	CA-2015-1	########	########	Standard C	BH-11710	Brosina Ho	Consumer	United Sta	Los Angele	California	90032	West	OFF-AR-1(	Office Sup	Art	Newell 32	7.28
9	8	CA-2015-1	########	########	Standard C	BH-11710	Brosina Ho	Consumer	United Sta	Los Angele	California	90032	West	TEC-PH-1(	Technolog	Phones	Mitel 5320	907.152
10	9	CA-2015-1	########	########	Standard C	BH-11710	Brosina Ho	Consumer	United Sta	Los Angele	California	90032	West	OFF-BI-10	Office Sup	Binders	DXL Angle-	18.504
11	10	CA-2015-1	########	########	Standard C	BH-11710	Brosina Ho	Consumer	United Sta	Los Angele	California	90032	West	OFF-AP-10	Office Sup	Appliances	Belkin F5C	114.9
12	11	CA-2015-1	########	########	Standard C	BH-11710	Brosina Ho	Consumer	United Sta	Los Angele	California	90032	West	FUR-TA-1(	Furniture	Tables	Chromcraf	1706.184
13	12	CA-2015-1	########	########	Standard C	BH-11710	Brosina Ho	Consumer	United Sta	Los Angele	California	90032	West	TEC-PH-1(	Technolog	Phones	Konftel 25	911.424
14	13	CA-2018-1	########	########	Standard C	AA-10480	Andrew All	Consumer	United Sta	Concord	North Car	28027	South	OFF-PA-10	Office Sup	Paper	Xerox 196	15.552
15	14	CA-2017-1	########	########	Standard C	IM-15070	Irene Mad	Consumer	United Sta	Seattle	Washingto	98103	West	OFF-BI-10	Office Sup	Binders	Fellowes P	407.976
16	15	US-2016-1	########	########	Standard C	HP-14815	Harold Pa	Home Offi	United Sta	Fort Wortl	Texas	76106	Central	OFF-AP-10	Office Sup	Appliances	Holmes Re	68.81
17	16	US-2016-1	########	########	Standard C	HP-14815	Harold Pa	Home Offi	United Sta	Fort Wortl	Texas	76106	Central	OFF-BI-10	Office Sup	Binders	Storex Dur	2.544
18	17	CA-2015-1	########	########	Standard C	PK-19075	Pete Kriz	Consumer	United Sta	Madison	Wisconsin	53711	Central	OFF-ST-10	Office Sup	Storage	Stur-D-Sto	665.88
19	18	CA-2015-1	########	########	Second Cla	AG-10270	Alejandro	Consumer	United Sta	West Jordi	Utah	84084	West	OFF-ST-10	Office Sup	Storage	Fellowes S	55.5
20	19	CA-2015-1	########	########	Second Cla	ZD-21925	Zuschuss D	Consumer	United Sta	San Franci	California	94109	West	OFF-AR-1(	Office Sup	Art	Newell 34	8.56
21	20	CA-2015-1	########	########	Second Cla	ZD-21925	Zuschuss D	Consumer	United Sta	San Franci	California	94109	West	TEC-PH-1(	Technolog	Phones	Cisco SPA	213.48
22	21	CA-2015-1	########	########	Second Cla	ZD-21925	Zuschuss D	Consumer	United Sta	San Franci	California	94109	West	OFF-BI-10	Office Sup	Binders	Wilson Jor	22.72
23	22	CA-2017-1	########	########	Standard C	KB-16585	Ken Black	Corporate	United Sta	Fremont	Nebraska	68025	Central	OFF-AR-1(	Office Sup	Art	Newell 31	19.46
24	23	CA-2017-1	########	########	Standard C	KB-16585	Ken Black	Corporate	United Sta	Fremont	Nebraska	68025	Central	OFF-AP-10	Office Sup	Appliances	Acco Six-C	60.34
25	24	US-2018-1	########	########	Second Cla	SF-20065	Sandra Fla	Consumer	United Sta	Philadelph	Pennsylvar	19140	East	FUR-CH-1(	Furniture	Chairs	Global Del	71.372
26	25	CA-2016-1	########	########	Standard C	EB-13870	Emily Burn	Consumer	United Sta	Orem	Utah	84057	West	FUR-TA-1(	Furniture	Tables	Bretford C	1044.63
27	26	CA-2017-1	########	########	Second Cla	EH-13945	Eric Hoffm	Consumer	United Sta	Los Angele	California	90049	West	OFF-BI-10	Office Sup	Binders	Wilson Jor	11.648

Figure 1.1(uncleaned data)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
9775	9774	CA-2017-1	########	########	Standard C	PF-19225	Phillip Flat	Consumer	United Sta	Atlanta	Georgia	30318	South	TEC-PH-10	Technolog	Phones	Cisco IPÅ f	135.95
9776	9775	CA-2015-1	########	########	Standard C	LF-17185	Luke Foste	Consumer	United Sta	San Anton	Texas	78207	Central	OFF-BI-10	Office Sup	Binders	GBC Docul	2177.584
9777	9776	CA-2015-1	########	########	Standard C	LF-17185	Luke Foste	Consumer	United Sta	San Anton	Texas	78207	Central	FUR-FU-1	Furniture	Furnishing	DAX Clear	17.496
9778	9777	CA-2015-1	########	########	Standard C	LF-17185	Luke Foste	Consumer	United Sta	San Anton	Texas	78207	Central	OFF-BI-10	Office Sup	Binders	GBC Premi	16.784
9779	9778	CA-2015-1	########	########	Standard C	LF-17185	Luke Foste	Consumer	United Sta	San Anton	Texas	78207	Central	TEC-AC-1	Technolog	Accessorie	Microsoft	431.136
9780	9779	CA-2015-1	########	########	Standard C	LF-17185	Luke Foste	Consumer	United Sta	San Anton	Texas	78207	Central	OFF-BI-10	Office Sup	Binders	GBC Instar	8.88
9781	9780	CA-2015-1	########	########	Standard C	LF-17185	Luke Foste	Consumer	United Sta	San Anton	Texas	78207	Central	OFF-AP-10	Office Sup	Appliances	Acco 6 Ou	4.836
9782	9781	CA-2017-1	########	########	Standard C	CL-12565	Clay Ludtk	Consumer	United Sta	Long Beac	New York	11561	East	TEC-PH-1	Technolog	Phones	Wi-Ex zBo	437.85
9783	9782	CA-2017-1	########	########	Standard C	CL-12565	Clay Ludtk	Consumer	United Sta	Long Beac	New York	11561	East	OFF-BI-10	Office Sup	Binders	GBC Docul	673.568
9784	9783	CA-2016-1	########	########	Second Cla	EM-13825	Elizabeth I	Corporate	United Sta	Paterson	New Jerse	7501	East	OFF-PA-10	Office Sup	Paper	Xerox 189	274.8
9785	9784	CA-2016-1	########	########	Second Cla	EM-13825	Elizabeth I	Corporate	United Sta	Paterson	New Jerse	7501	East	OFF-ST-10	Office Sup	Storage	Fellowes N	62.18
9786	9785	CA-2016-1	########	########	Second Cla	EM-13825	Elizabeth I	Corporate	United Sta	Paterson	New Jerse	7501	East	FUR-FU-1	Furniture	Furnishing	Eldon Imag	8.28
9787	9786	CA-2016-1	########	########	Standard C	ME-17725	Max Engle	Consumer	United Sta	Louisville	Kentucky	40214	South	OFF-BI-10	Office Sup	Binders	Acco Flexil	48.81
9788	9787	US-2015-1	########	########	Same Day	BG-11035	Barry Gon	Consumer	United Sta	Hampton	Virginia	23666	South	FUR-CH-1	Furniture	Chairs	Global Sta	149.9
9789	9788	CA-2018-1	########	########	Standard C	CJ-12010	Caroline Ju	Consumer	United Sta	Houston	Texas	77070	Central	FUR-BO-1	Furniture	Bookcases	Atlantic M	1023.332
9790	9789	CA-2018-1	########	########	Standard C	CJ-12010	Caroline Ju	Consumer	United Sta	Houston	Texas	77070	Central	FUR-CH-1	Furniture	Chairs	Global Del	600.558
9791	9790	CA-2018-1	########	########	Standard C	CJ-12010	Caroline Ju	Consumer	United Sta	Houston	Texas	77070	Central	TEC-AC-1	Technolog	Accessorie	Kensingtor	39.992
9792	9791	CA-2018-1	########	########	Standard C	CJ-12010	Caroline Ju	Consumer	United Sta	Houston	Texas	77070	Central	FUR-CH-1	Furniture	Chairs	Global Lea	211.246
9793	9792	CA-2015-1	########	########	Second Cla	KH-16360	Katherine	Consumer	United Sta	Houston	Texas	77070	Central	OFF-EN-1	Office Sup	Envelopes	Staple env	56.064
9794	9793	CA-2015-1	########	########	Second Cla	KH-16360	Katherine	Consumer	United Sta	Houston	Texas	77070	Central	FUR-CH-1	Furniture	Chairs	Global Del	107.772
9795	9794	CA-2015-1	########	########	Second Cla	KH-16360	Katherine	Consumer	United Sta	Houston	Texas	77070	Central	OFF-PA-10	Office Sup	Paper	Adams Tel	4.832
9796	9795	CA-2015-1	########	########	Second Cla	KH-16360	Katherine	Consumer	United Sta	Houston	Texas	77070	Central	OFF-BI-10	Office Sup	Binders	Ibico Plast	18.24
9797	9796	CA-2017-1	########	########	Standard C	SH-19975	Sally Hugh	Corporate	United Sta	Chicago	Illinois	60610	Central	OFF-BI-10	Office Sup	Binders	Cardinal H	3.798
9798	9797	CA-2016-1	########	########	Standard C	CS-12490	Cindy Schr	Corporate	United Sta	Toledo	Ohio	43615	East	OFF-AR-1	Office Sup	Art	BIC Brite L	10.368
9799	9798	CA-2016-1	########	########	Standard C	CS-12490	Cindy Schr	Corporate	United Sta	Toledo	Ohio	43615	East	TEC-PH-1	Technolog	Phones	GE 30524E	235.188
9800	9799	CA-2016-1	########	########	Standard C	CS-12490	Cindy Schr	Corporate	United Sta	Toledo	Ohio	43615	East	TEC-PH-1	Technolog	Phones	Anker 24W	26.376
9801	9800	CA-2016-1	########	########	Standard C	CS-12490	Cindy Schr	Corporate	United Sta	Toledo	Ohio	43615	East	TEC-AC-1	Technolog	Accessorie	SanDisk Cr	10.384

Fig 1.2(uncleaned data)

	Row ID	Order ID	Order Date	Month	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
2	1	CA-2017-152156	08-11-2017	Nov	11-11-2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase
3	2	CA-2017-152156	08-11-2017	Nov	11-11-2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Ro
4	3	CA-2017-138688	12-06-2017	Jun	16-06-2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by
5	4	US-2016-108966	11-10-2016	Oct	18-10-2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular T
6	5	US-2016-108966	11-10-2016	Oct	18-10-2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System
7	6	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories
8	7	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322
9	8	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone
10	9	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by
11	10	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge
12	11	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tabl
13	12	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 ConferenceÅ phoneÅ - Charcoal
14	13	CA-2018-114412	15-04-2018	Apr	20-04-2018	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	orth Carolin	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967
15	14	CA-2017-161389	05-12-2017	Dec	10-12-2017	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Mach
16	15	US-2016-118983	22-11-2016	Nov	26-11-2016	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances	mes Replacement Filter for HEPA Air Cleaner, Very Larg
17	16	US-2016-118983	22-11-2016	Nov	26-11-2016	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10000756	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Bi
18	17	CA-2015-105893	11-11-2015	Nov	18-11-2015	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-10004186	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"W
19	18	CA-2015-167164	13-05-2015	May	15-05-2015	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Drawer
20	19	CA-2015-143336	27-08-2015	Aug	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10003056	Office Supplies	Art	Newell 341
21	20	CA-2015-143336	27-08-2015	Aug	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone
22	21	CA-2015-143336	27-08-2015	Aug	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215	Office Supplies	Binders	Wilson Jones Hanging View Binder, White
23	22	CA-2017-137330	09-12-2017	Dec	13-12-2017	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10000246	Office Supplies	Art	Newell 318
24	23	CA-2017-137330	09-12-2017	Dec	13-12-2017	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AP-10001492	Office Supplies	Appliances	Acco Six-Outlet Power Strip, 4' Cord Leng
25	24	US-2018-156909	16-07-2018	Jul	18-07-2018	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	ennsylvania	19140	East	FUR-CH-10002774	Furniture	Chairs	Global Deluxe Stacking Chair, Gray
26	25	CA-2016-106320	25-09-2016	Sep	30-09-2016	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular T
27	26	CA-2017-121755	16-01-2017	Jan	20-01-2017	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001634	Office Supplies	Binders	Wilson Jones Active Use Binders
28	27	CA-2017-121755	16-01-2017	Jan	20-01-2017	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories	ImationÅ 8GB Mini TravelDrive USB 2.0Å Flas
29	28	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	ennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palais Royal Lawyers Bookcase, Royale C
30	29	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	ennsylvania	19140	East	OFF-BI-10000474	Office Supplies	Binders	Avery Recycled Flexi-View Covers for Binding S
31	30	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	ennsylvania	19140	East	FUR-FU-10004848	Furniture	Furnishings	Howard Miller 13-3/4" Diameter Brushed Chrome Ro
32	31	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	ennsylvania	19140	East	OFF-EN-10001509	Office Supplies	Envelopes	Poly String Tie Envelopes
33	32	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	ennsylvania	19140	East	OFF-AR-10004042	Office Supplies	Art	BOSTON Model 1800 Electric Pencil Sharpeners, Put
34	33	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	ennsylvania	19140	East	OFF-BI-10001525	Office Supplies	Binders	cco Pressboard Covers with Storage Hooks, 14 7/8" x 1
35	34	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	ennsylvania	19140	East	OFF-AR-10001683	Office Supplies	Art	Lumber Crayons
36	35	CA-2018-107727	19-10-2018	Oct	23-10-2018	Second Class	MA-17560	Matt Abelman	Home Office	United States	Houston	Texas	77095	Central	OFF-PA-10000249	Office Supplies	Paper	Easy-staple paper
37	36	CA-2017-117590	08-12-2017	Dec	10-12-2017	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones	GE 30524EE4
38	37	CA-2017-117590	08-12-2017	Dec	10-12-2017	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	FUR-FU-10003664	Furniture	Furnishings	Electrix Architect's Clamp-On Swing Arm Lamp
39	38	CA-2016-117415	27-12-2016	Dec	31-12-2016	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central	OFF-EN-10002986	Office Supplies	Envelopes	#10-4 1/8" x 9 1/2" Premium Diagonal Seam E

Fig 2.1(Cleaned Data)

2961	2960	CA-2016-158456	24-12-2016	Dec	29-12-2016	Standard Class	KT-16465	Kean Takahito	Consumer	United States	Los Angeles	California	90032	West	OFF-FA-10004854	Office Supplies	Fasteners	Vinyl Coated Wire Paper Clips in Organizer Box
2962	2961	CA-2018-143665	26-04-2018	Apr	01-05-2018	Standard Class	PO-19180	Phillisse Overcash	Home Office	United States	Seattle	Washington	98115	West	OFF-PA-10003673	Office Supplies	Paper	Strathmore Photo Mount Cards
2963	2962	CA-2018-143665	26-04-2018	Apr	01-05-2018	Standard Class	PO-19180	Phillisse Overcash	Home Office	United States	Seattle	Washington	98115	West	OFF-LA-10002034	Office Supplies	Labels	Avery 478
2964	2963	CA-2018-137428	16-12-2018	Dec	21-12-2018	Second Class	AY-10555	Andy Yotov	Corporate	United States	Oceanside	California	92054	West	FUR-CH-10002774	Furniture	Chairs	Global Deluxe Stacking Chair, Gray
2965	2964	CA-2018-137428	16-12-2018	Dec	21-12-2018	Second Class	AY-10555	Andy Yotov	Corporate	United States	Oceanside	California	92054	West	FUR-CH-10003817	Furniture	Chairs	Global Value Steno Chair, Gray
2966	2965	CA-2018-137428	16-12-2018	Dec	21-12-2018	Second Class	AY-10555	Andy Yotov	Corporate	United States	Oceanside	California	92054	West	OFF-BI-10002949	Office Supplies	Binders	Prestige Round Ring Binders
2967	2966	CA-2018-137428	16-12-2018	Dec	21-12-2018	Second Class	AY-10555	Andy Yotov	Corporate	United States	Oceanside	California	92054	West	FUR-FU-10002445	Furniture	Furnishings	DAX Two-Tone Rosewood/Black Document Frame,
2968	2967	CA-2015-162866	27-12-2015	Dec	31-12-2015	Standard Class	Co-12640	Corey-Lock	Consumer	United States	Skokie	Illinois	60076	Central	FUR-FU-10001473	Furniture	Furnishings	DAX Wood Document Frame
2969	2968	CA-2015-162866	27-12-2015	Dec	31-12-2015	Standard Class	Co-12640	Corey-Lock	Consumer	United States	Skokie	Illinois	60076	Central	OFF-ST-10002562	Office Supplies	Storage	Staple magnet
2970	2969	CA-2018-167941	06-11-2018	Nov	09-11-2018	Second Class	JF-15565	Jill Fjeld	Consumer	United States	Fayetteville	orth Carolin	28314	South	OFF-AP-10002118	Office Supplies	Appliances	1.7 Cubic Foot Compact "Cube" Office Refri
2971	2970	CA-2018-167941	06-11-2018	Nov	09-11-2018	Second Class	JF-15565	Jill Fjeld	Consumer	United States	Fayetteville	orth Carolin	28314	South	OFF-PA-10001800	Office Supplies	Paper	Xerox 220
2972	2971	CA-2018-167941	06-11-2018	Nov	09-11-2018	Second Class	JF-15565	Jill Fjeld	Consumer	United States	Fayetteville	orth Carolin	28314	South	OFF-BI-10002827	Office Supplies	Binders	Avery Durable Poly Binders
2973	2972	CA-2018-167941	06-11-2018	Nov	09-11-2018	Second Class	JF-15565	Jill Fjeld	Consumer	United States	Fayetteville	orth Carolin	28314	South	FUR-FU-10004671	Furniture	Furnishings	Executive Impressions 12" Wall Cloc
2974	2973	CA-2018-167941	06-11-2018	Nov	09-11-2018	Second Class	JF-15565	Jill Fjeld	Consumer	United States	Fayetteville	orth Carolin	28314	South	OFF-ST-10000736	Office Supplies	Storage	Carina Double Wide Media Storage Towers in Na
2975	2974	CA-2018-111808	16-12-2018	Dec	20-12-2018	Standard Class	AR-10510	Andrew Roberts	Consumer	United States	Tulsa	Oklahoma	74133	Central	OFF-BI-10004656	Office Supplies	Binders	Peel & Stick Add-On Corner Pockets
2976	2975	CA-2016-137512	07-05-2016	May	12-05-2016	Standard Class	AG-10675	Anna Gayman	Consumer	United States	Allen	Texas	75002	Central	FUR-TA-10001095	Furniture	Tables	Chromcraft Round Conference Table
2977	2976	CA-2016-137512	07-05-2016	May	12-05-2016	Standard Class	AG-10675	Anna Gayman	Consumer	United States	Allen	Texas	75002	Central	OFF-PA-10000213	Office Supplies	Paper	Xerox 198
2978	2977	CA-2018-139773	04-12-2018	Dec	04-12-2018	Same Day	DV-13045	Darrin Van Huff	Corporate	United States	Philadelphia	ennsylvania	19143	East	FUR-CH-10001797	Furniture	Chairs	Safco Chair Connectors, 6/Carton
2979	2978	CA-2018-134607	03-10-2018	Oct	08-10-2018	Second Class	HL-15040	Hunter Lopez	Consumer	United States	Rochester	New York	14609	East	OFF-ST-10002214	Office Supplies	Storage	X-Rack File for Hanging Folders
2980	2979	CA-2015-109232	13-01-2015	Jan	16-01-2015	Second Class	ND-18370	Natalie DeCherney	Consumer	United States	Mount Pleasant	outh Carolin	29464	South	FUR-CH-10000422	Furniture	Chairs	Global Highback Leather Tilter in Burgu
2981	2980	CA-2016-129850	12-06-2016	Jun	17-06-2016	Standard Class	GB-14575	Giulietta Baptist	Consumer	United States	Philadelphia	ennsylvania	19134	East	OFF-PA-10003848	Office Supplies	Paper	Xerox 1997
2982	2981	CA-2016-129850	12-06-2016	Jun	17-06-2016	Standard Class	GB-14575	Giulietta Baptist	Consumer	United States	Philadelphia	ennsylvania	19134	East	FUR-FU-10003623	Furniture	Furnishings	DataProducts Ampli Magnifier Task Lamp,
2983	2982	CA-2015-131310	12-07-2015	Jul	18-07-2015	Standard Class	CL-12565	Clay Ludtke	Consumer	United States	Seattle	Washington	98115	West	FUR-CH-10001797	Furniture	Chairs	Safco Chair Connectors, 6/Carton
2984	2983	CA-2015-131310	12-07-2015	Jul	18-07-2015	Standard Class	CL-12565	Clay Ludtke	Consumer	United States	Seattle	Washington	98115	West	OFF-BI-10003094	Office Supplies	Binders	Self-Adhesive Ring Binder Labels
2985	2984	US-2015-112872	06-12-2015	Dec	11-12-2015	Second Class	RC-19960	Ryan Crowe	Consumer	United States	Springfield	Oregon	97477	West	OFF-ST-10002205	Office Supplies	Storage	File Shuttle I and Handi-File
2986	2985	US-2015-112872	06-12-2015	Dec	11-12-2015	Second Class	RC-19960	Ryan Crowe	Consumer	United States	Springfield	Oregon	97477	West	FUR-TA-10003238	Furniture	Tables	Chromcraft Bull-Nose Wood 48" x 96" Rectangular Co
2987	2986	CA-2017-139269	25-05-2017	May	30-05-2017	Standard Class	JB-16045	Julia Barnett	Home Office	United States	Columbus	Georgia	31907	South	FUR-FU-10000755	Furniture	Furnishings	Eldon Expressions Mahogany Wood Desk Co
2988	2987	CA-2017-139269	25-05-2017	May	30-05-2017	Standard Class	JB-16045	Julia Barnett	Home Office	United States	Columbus	Georgia	31907	South	OFF-PA-10002893	Office Supplies	Paper	Wirebound Service Call Books, 5 1/2" x
2989	2988	CA-2017-139269	25-05-2017	May	30-05-2017	Standard Class	JB-16045	Julia Barnett	Home Office	United States	Columbus	Georgia	31907	South	OFF-ST-10000689	Office Supplies	Storage	Fellowes Strictly Business Drawer File, Letter/
2990	2989	CA-2017-139010	21-02-2017	Feb	26-02-2017	Second Class	MC-17635	Matthew Clasen	Corporate	United States	Los Angeles	California	90032	West	TEC-AC-10004227	Technology	Accessories	SanDisk Ultra 16 GB MicroSDHC Class 10 Men
2991	2990	CA-2017-139010	21-02-2017	Feb	26-02-2017	Second Class	MC-17635	Matthew Clasen	Corporate	United States	Los Angeles	California	90032	West	OFF-BI-10000174	Office Supplies	Binders	Wilson Jones Clip & Carry Folder Binder Tool for Rin
2992	2991	CA-2017-139010	21-02-2017	Feb	26-02-2017	Second Class	MC-17635	Matthew Clasen	Corporate	United States	Los Angeles	California	90032	West	OFF-ST-10001809	Office Supplies	Storage	Fellowes Officeware Wire Shelving
2993	2992	CA-2017-139010	21-02-2017	Feb	26-02-2017	Second Class	MC-17635	Matthew Clasen	Corporate	United States	Los Angeles	California	90032	West	OFF-BI-10002309	Office Supplies	Binders	Avery Heavy-Duty EZD Binder With Locking
2994	2993	CA-2017-134376	02-10-2017	Oct	06-10-2017	Standard Class	TT-21265	Tim Taslimi	Corporate	United States	New York City	New York	10024	East	OFF-AP-10004532	Office Supplies	Appliances	Kensington 6 Outlet Guardian Standard Surge
2995	2994	CA-2018-147291	11-03-2018	Mar	17-03-2018	Standard Class	MJ-17740	Max Jones	Consumer	United States	Detroit	Michigan	48227	Central	OFF-BI-10003091	Office Supplies	Binders	GBC DocuBind TL200 Manual Binding Ma
2996	2995	CA-2017-112893	09-09-2017	Sep	13-09-2017	Second Class	AT-10735	Annie Thurman	Consumer	United States	Stockton	California	95207	West	OFF-BI-10004654	Office Supplies	Binders	VariCap6 Expandable Binder
2997	2996	US-2015-150532	14-07-2015	Jul	21-07-2015	Standard Class	PB-19150	Philip Brown	Consumer	United States	Phoenix	Arizona	85023	West	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System
2998	2997	CA-2015-138317	21-06-2015	Jun	25-06-2015	Standard Class	NW-18400	Natalie Webber	Consumer	United States	Philadelphia	ennsylvania	19120	East	OFF-EN-10001539	Office Supplies	Envelopes	Staple envelope
2999	2998	CA-2015-138317	21-06-2015	Jun	25-06-2015	Standard Class	NW-18400	Natalie Webber	Consumer	United States	Philadelphia	ennsylvania	19120	East	FUR-FU-10000550	Furniture	Furnishings	Stacking Trays by OIC

Fig 2.2(Cleaned data)

4.2 Visualization / Output Results

Chart 1 – *Monthly Sales Trend*

Line chart showing sales over time.

Chart 2 – *Sales by Subcategory*

Bar chart comparing sales in different subcategories.

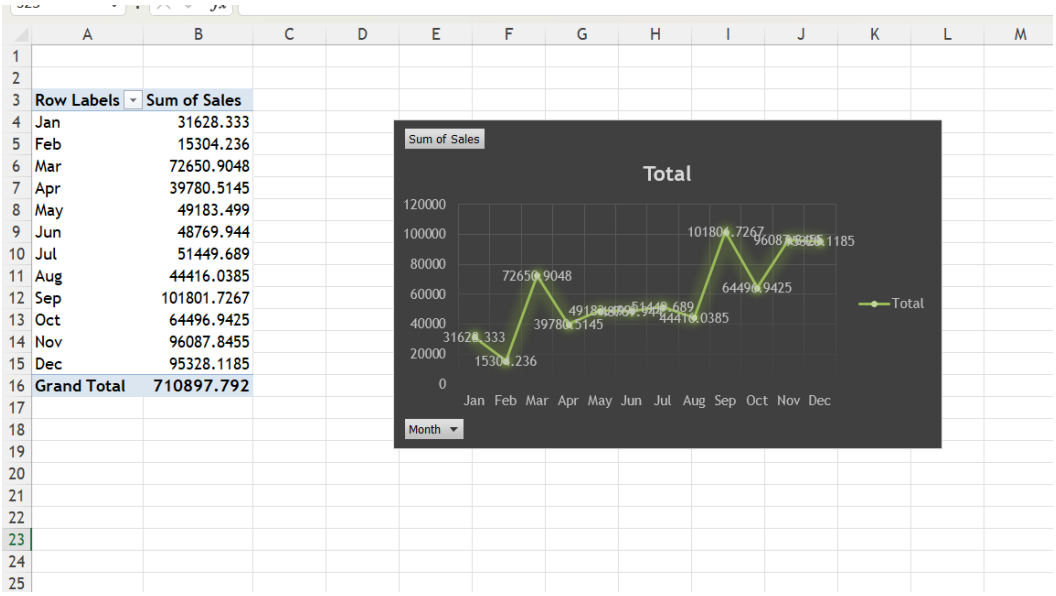
Chart 3 – *Sales by Category*

Column chart showing sales in different category.

Chart 4 – *Sales by region*

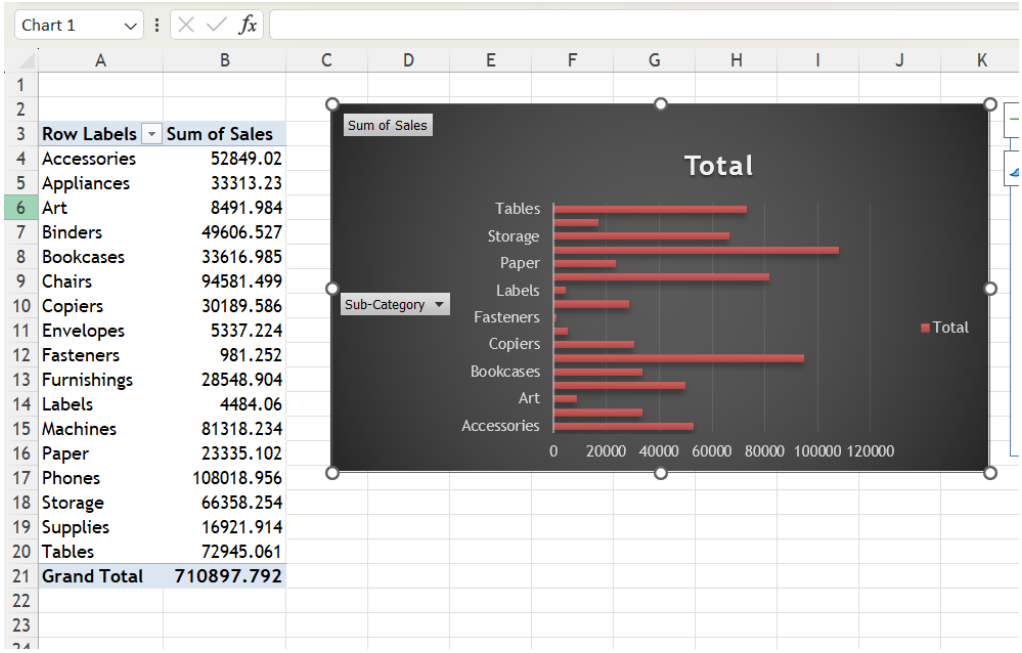
Pie chart showing sales in a region.

Fig -3 (Dashboard)



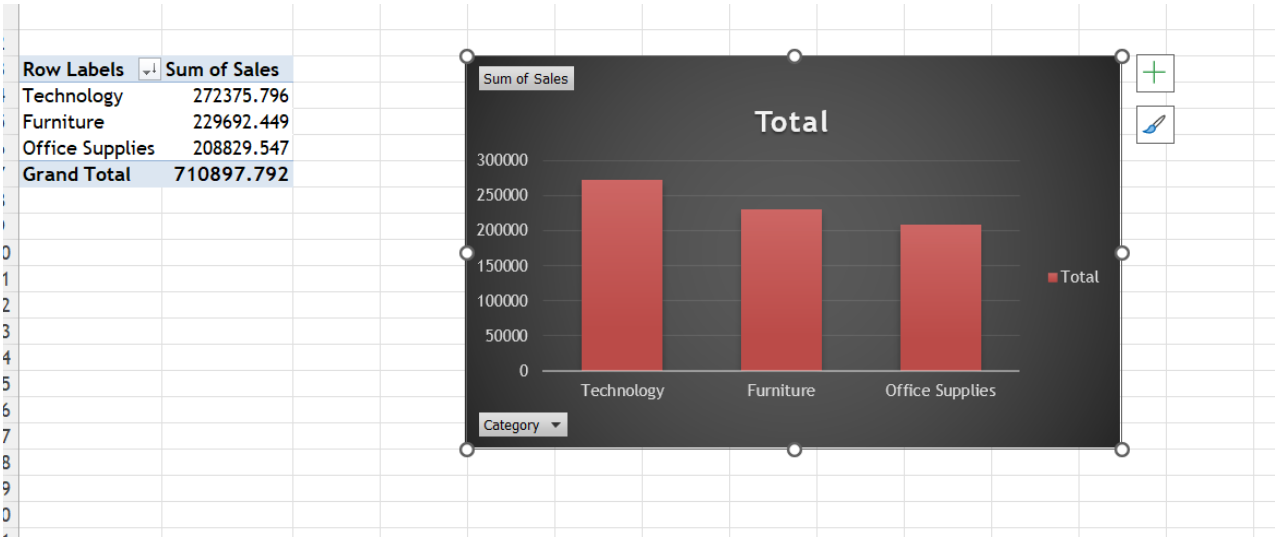
Row Labels	Sum of Sales
Jan	31628.333
Feb	15304.236
Mar	72650.9048
Apr	39780.5145
May	49183.499
Jun	48769.944
Jul	51449.689
Aug	44416.0385
Sep	101801.7267
Oct	64496.9425
Nov	96087.8455
Dec	95328.1185
Grand Total	710897.792

Fig- 4.1(Line Chart)



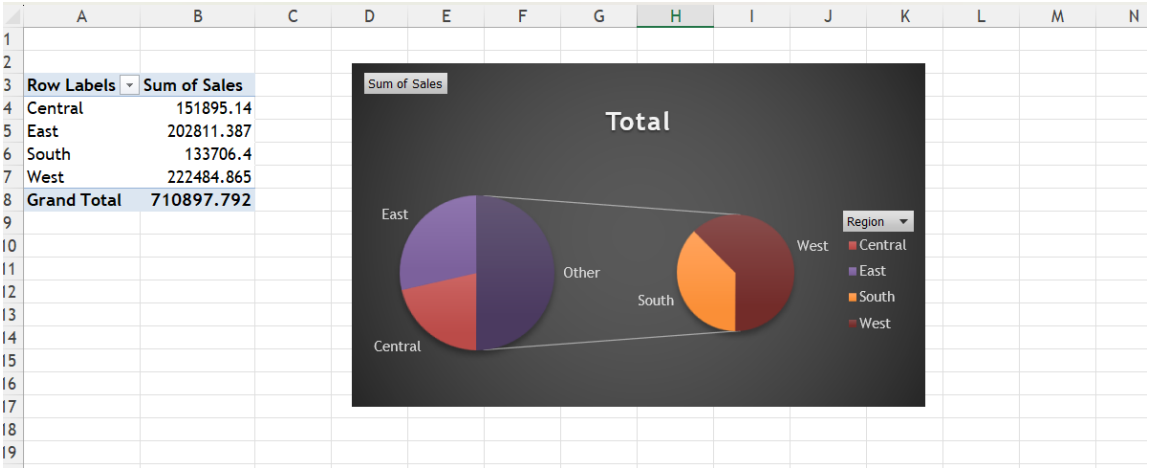
Row Labels	Sum of Sales
Accessories	52849.02
Appliances	33313.23
Art	8491.984
Binders	49606.527
Bookcases	33616.985
Chairs	94581.499
Copiers	30189.586
Envelopes	5337.224
Fasteners	981.252
Furnishings	28548.904
Labels	4484.06
Machines	81318.234
Paper	23335.102
Phones	108018.956
Storage	66358.254
Supplies	16921.914
Tables	72945.061
Grand Total	710897.792

Fig 4.2 (Bar Chart)



Row Labels	Sum of Sales
Technology	272375.796
Furniture	229692.449
Office Supplies	208829.547
Grand Total	710897.792

Fig 4.3(Column Chart)



Row Labels	Sum of Sales
Central	151895.14
East	202811.387
South	133706.4
West	222484.865
Grand Total	710897.792

Fig 4.4 (Pie Chart)

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- Some regions generate higher sales than others.
- Certain product categories produce higher profits.
- Customer payment preferences vary.

5.2 Project Links

Kaggle: https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting

5.3 Limitations

- Dataset is limited to sample data.
- No real-time updates.
- Advanced machine learning models were not used.

5.4 Future Work

Possible improvements:

- Apply machine learning for sales forecasting.
- Integrate real-time business data.
- Build web-based dashboards.

5.5 Conclusion

This project demonstrated how retail sales data can be cleaned and analyzed using Excel. By creating visual dashboards, the project helps identify business patterns such as sales trends and product performance.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

1	Row ID	Order ID	Order Date	Month	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
2	1	CA-2017-152156	08-11-2017	Nov	11-11-2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase
3	2	CA-2017-152156	08-11-2017	Nov	11-11-2017	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs, Ro
4	3	CA-2017-138688	12-06-2017	Jun	16-06-2017	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters by
5	4	US-2016-108966	11-10-2016	Oct	18-10-2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular T
6	5	US-2016-108966	11-10-2016	Oct	18-10-2016	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System
7	6	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Accessories
8	7	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322
9	8	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone
10	9	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by
11	10	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge
12	11	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tabl
13	12	CA-2015-115812	09-06-2015	Jun	14-06-2015	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 ConferenceÃ phoneÃ - Charcoal
14	13	CA-2018-114412	15-04-2018	Apr	20-04-2018	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	orth Carolin	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967
15	14	CA-2017-161389	05-12-2017	Dec	10-12-2017	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Mach
16	15	US-2016-118983	22-11-2016	Nov	26-11-2016	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances	mes Replacement Filter for HEPA Air Cleaner, Very Larg
17	16	US-2016-118983	22-11-2016	Nov	26-11-2016	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10000756	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Bi
18	17	CA-2015-105893	11-11-2015	Nov	18-11-2015	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-10004186	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 36"V
19	18	CA-2015-167164	13-05-2015	May	15-05-2015	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Drawer
20	19	CA-2015-143336	27-08-2015	Aug	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10003056	Office Supplies	Art	Newell 341
21	20	CA-2015-143336	27-08-2015	Aug	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone
22	21	CA-2015-143336	27-08-2015	Aug	01-09-2015	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215	Office Supplies	Binders	Wilson Jones Hanging View Binder, White
23	22	CA-2017-137330	09-12-2017	Dec	13-12-2017	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10000246	Office Supplies	Art	Newell 318
24	23	CA-2017-137330	09-12-2017	Dec	13-12-2017	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AP-10001492	Office Supplies	Appliances	Acco Six-Outlet Power Strip, 4' Cord Leng
25	24	US-2018-156909	16-07-2018	Jul	18-07-2018	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-10002774	Furniture	Chairs	Global Deluxe Stacking Chair, Gray
26	25	CA-2016-106320	25-09-2016	Sep	30-09-2016	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular T
27	26	CA-2017-121755	16-01-2017	Jan	20-01-2017	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001634	Office Supplies	Binders	Wilson Jones Active Use Binders
28	27	CA-2017-121755	16-01-2017	Jan	20-01-2017	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories	ImationÃ 8GB Mini TravelDrive USB 2.0Ã Flas
29	28	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palais Royal Lawyers Bookcase, Royale C
30	29	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-10000474	Office Supplies	Binders	Avery Recycled Flexi-View Covers for Binding S
31	30	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-FU-10004848	Furniture	Furnishings	Howard Miller 13-3/4" Diameter Brushed Chrome Ro
32	31	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-EN-10001509	Office Supplies	Envelopes	Poly String Tie Envelopes
33	32	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-AR-10004042	Office Supplies	Art	BOSTON Model 1800 Electric Pencil Sharpeners, Put
34	33	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-10001525	Office Supplies	Binders	ACCO Pressboard Covers with Storage Hooks, 14 7/8" x 1
35	34	US-2016-150630	17-09-2016	Sep	21-09-2016	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-AR-10001683	Office Supplies	Art	Lumber Crayons
36	35	CA-2018-107727	19-10-2018	Oct	23-10-2018	Second Class	MA-17560	Matt Abelman	Home Office	United States	Houston	Texas	77095	Central	OFF-PA-10000249	Office Supplies	Paper	Easy-staple paper
37	36	CA-2017-117590	08-12-2017	Dec	10-12-2017	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones	GE 30524EE4
38	37	CA-2017-117590	08-12-2017	Dec	10-12-2017	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	FUR-FU-10003664	Furniture	Furnishings	Electrix Architect's Clamp-On Swing Arm Lamp
39	38	CA-2016-117415	27-12-2016	Dec	31-12-2016	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central	OFF-EN-10002986	Office Supplies	Envelopes	#10-4 1/8" x 9 1/2" Premium Diagonal Seam E

Fig 6

The dataset used in this project contains retail sales transactions including transaction ID, date, customer information, product category, sales amount, and profit. The screenshot above shows a sample of the dataset used for analysis.

Appendix B — Key Excel Formulas Used

The following Excel tools and features were used during the project:

- **Pivot Tables** – Used to summarize sales data by region, product category, and time.
- **Sorting and Filtering** – Used to organize and examine specific parts of the dataset.
- **Remove Duplicates** – Used to clean the dataset and eliminate duplicate records.
- **Line Chart** – Used to visualize monthly sales trends.
- **Bar Chart** – Used to compare sales across different regions.
- **Pie Chart** – Used to show the distribution of payment methods.

These tools helped transform raw data into meaningful insights.